Week 1 Report - Medical Computer Vision Project

Project: Wound Type Classification, Detection, and Segmentation
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1) Week 1 Objective
Dataset selection, cleaning, and mini custom annotator development.

2) Course Scope Compliance
- Domain: Healthcare and medical imaging.
- Dataset source: Data in Brief journal (Elsevier).
- Pipeline targets: Classification, detection, segmentation.
- Custom annotation workflow included.

3) Dataset Details
- Name: Lower Limb and Feet Wound Image Dataset
- DOI: 10.1016/j.dib.2026.112730
- Download: https://data.mendeley.com/datasets/hsj38fwnvr/3
- License: CC BY 4.0
- Wound images: 2,686
- Normal images: 2,757
- Total images: 5,443
- Native image size: 331 x 331

4) Local Data Organization
- data/Nomal
- data/wound_main
- data/wound_mask

5) Week 1 Implementation Summary
- Script: week1/week1_dataset_clean.py
- Functions completed:
  a) Collect raw wound and normal images
  b) Validate and deduplicate images
  c) Normalize size and export to cleaned directories
  d) Export dataset manifest
  e) Select 20 sample images for annotation

6) Sampling and Cleaning Outcome
- Collected: 5,443 images
- Kept after cleaning: 5,265
- Label distribution in cleaned set:
  - miscellaneous (wound_main flat set): 2,508
  - normal: 2,757
- Sample manifest generated with 20 images and mixed labels.

7) Week 1 Deliverables Status
- Dataset proposal: completed
- Cleaning code: completed

- Custom annotator workflow: prepared
- 20 sample annotations: completed as CSV artifact

8) Rubric Alignment (Week 1-relevant)
- Dataset Selection (5%): addressed
- Annotator Development (10%): addressed
- Annotation Quality (5%): initial sample annotations prepared

9) Key Issue Encountered and Resolution
Issue: wound images were missed when assuming class subfolders.
Resolution: updated collection logic to read flat data/wound_main files and assign temporary 'miscellan

10) Plan for Week 2
- Complete full annotation pass.
- Train baseline classification model (ResNet/ViT).
- Report metrics and error analysis.

End of Week 1 Report